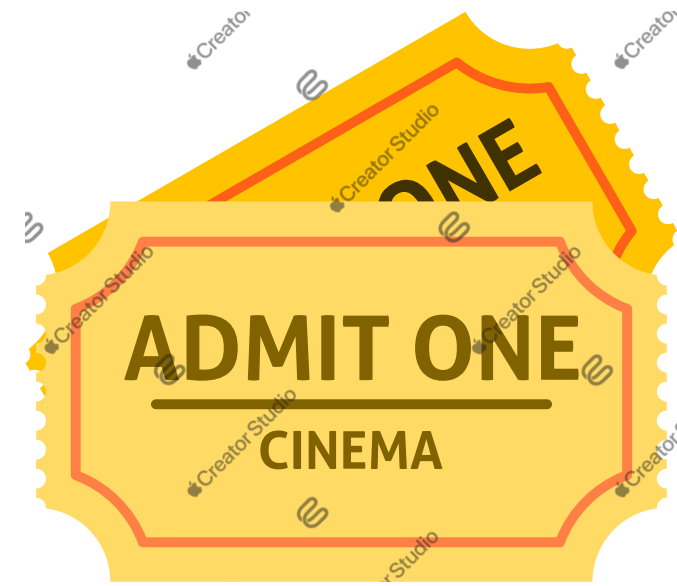


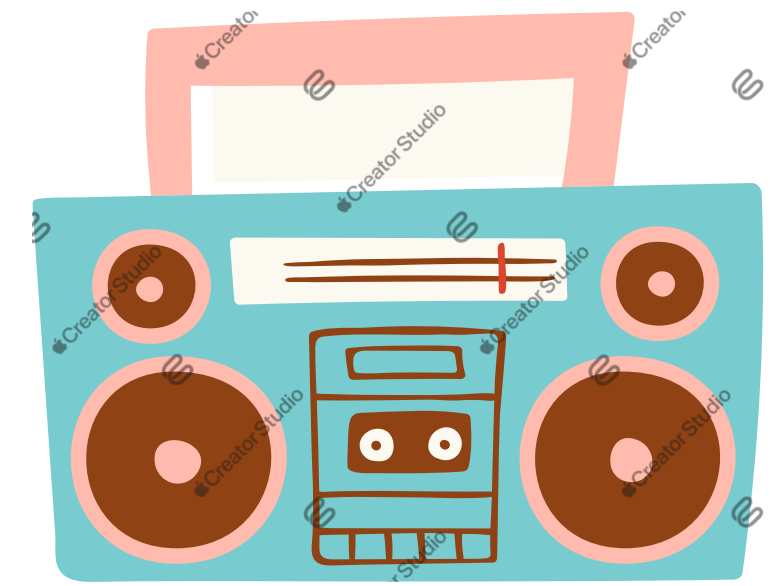
Canciones en la Pantalla

Grandes films, ¿grandes hits?



Motivación / Objetivos

Preguntas de investigación



- ¿Cómo se relaciona el éxito de una película con el éxito de su canción original?
- ¿Qué factores contribuyen al éxito de una canción de película?
- ¿Cuáles son las mejores canciones de película de todos los tiempos?



Criterio de Selección

Canción Original de Película

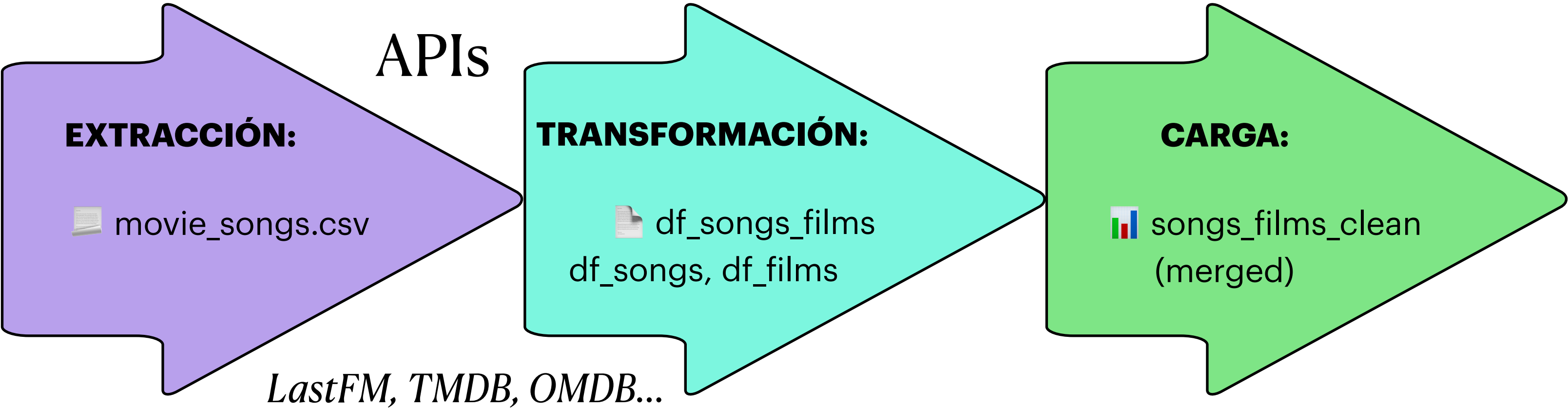


- Canciones con **letra y música originales**
- Compuestas específicamente para la película
- Debe haber claramente una presencia **audible e inteligible** tanto de la letra como la melodía, usadas a lo largo del **film**, o justo al principio de los **títulos de crédito**
- **NO** canciones preexistentes (ni versiones de canciones)

Metodología / Data Model

Extracción >> Transformación >> Carga

Fuentes:



Herramientas:



song_id	year	song_title	film_title	film_id
196212	1962	Under The Mango Tree	Dr. No	196212
196313	1963	From Russia With Love	From Russia With Love	196313
196413	1964	Goldfinger	Goldfinger	196413

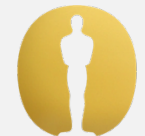
Metodología / Data Model

Extracción >> Transformación >> Carga

Fuentes:



WIKIPEDIA



THE ACADEMY
OF MOTION PICTURE ARTS AND SCIENCES



APIs

EXTRACCIÓN:

movie_songs.csv

TRANSFORMACIÓN:

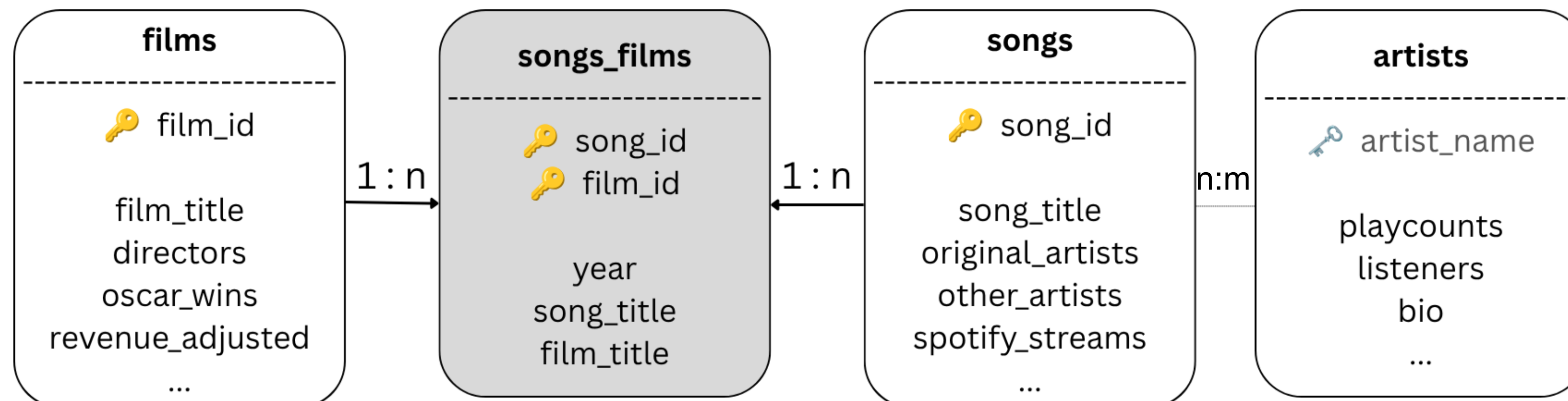
df_songs_films
df_songs, df_films

CARGA:

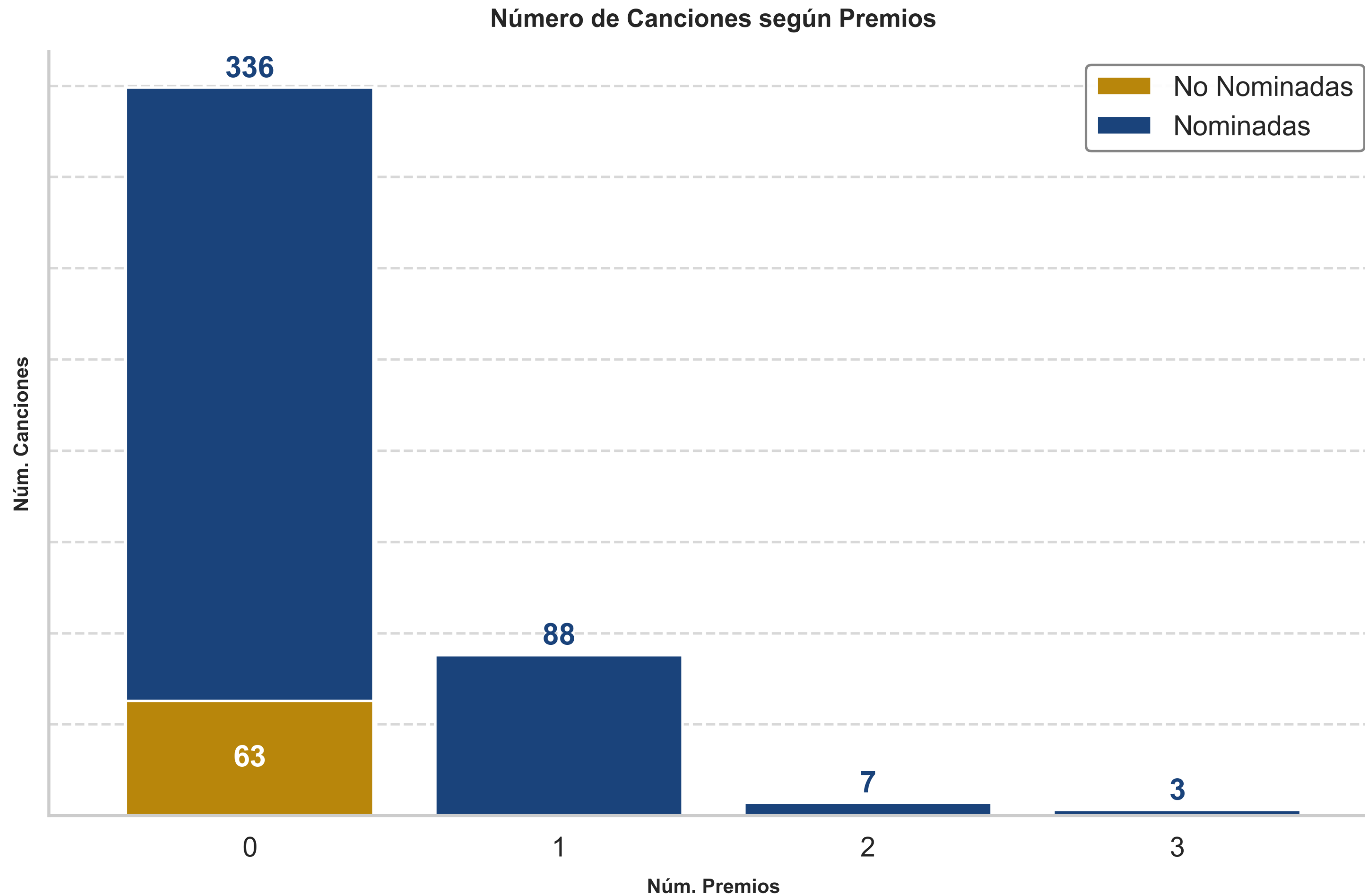
songs_films_clean
(merged)

LastFM, TMDB, OMDB...

Herramientas:

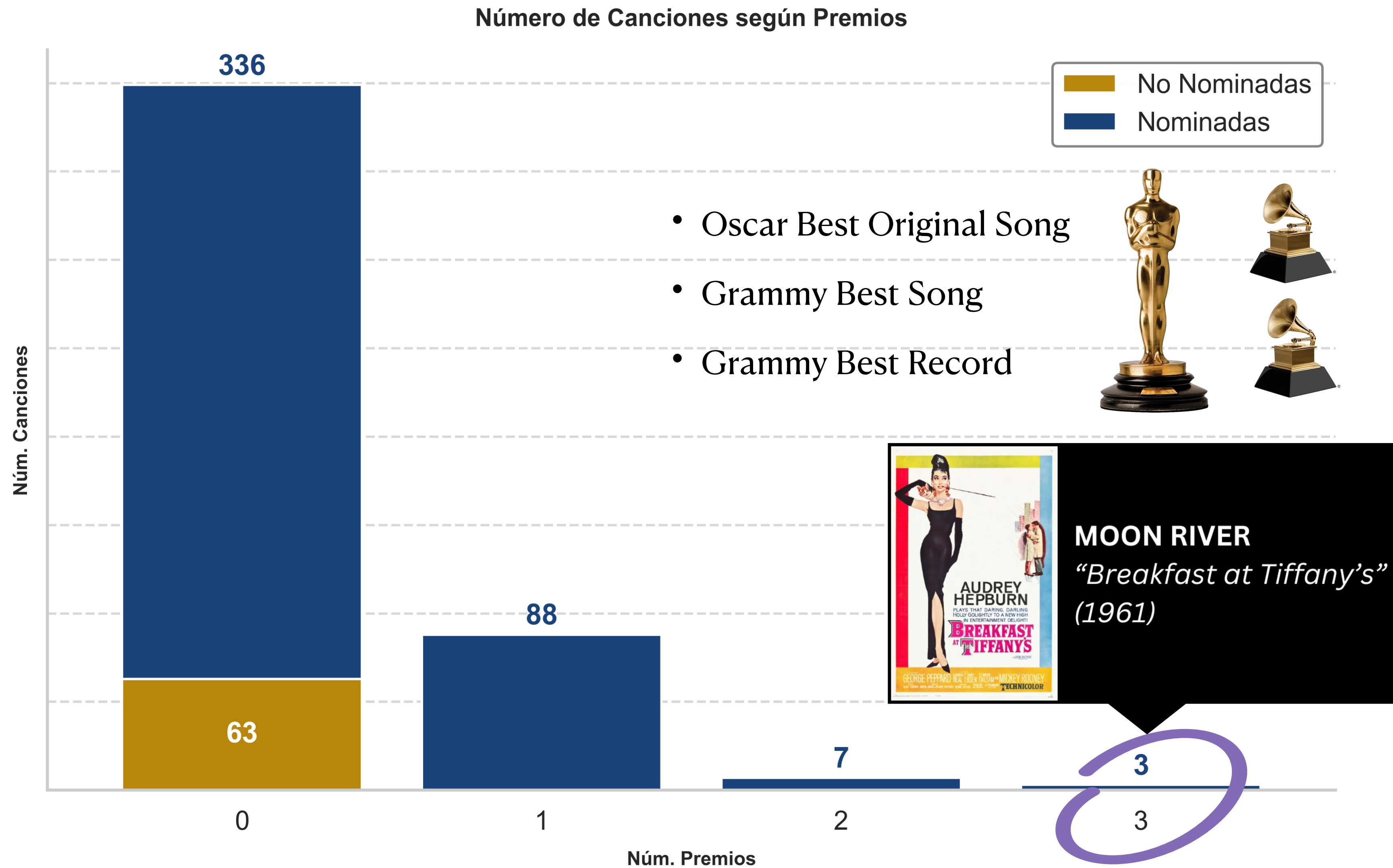


Exploración del Dataset



- 500 canciones
- 467 films
- 1937 -2025
- 602 artistas
- 2042 songwriters
- 196M streams (avg)
- 130 grabaciones (avg)
- 3'30" duración (avg)

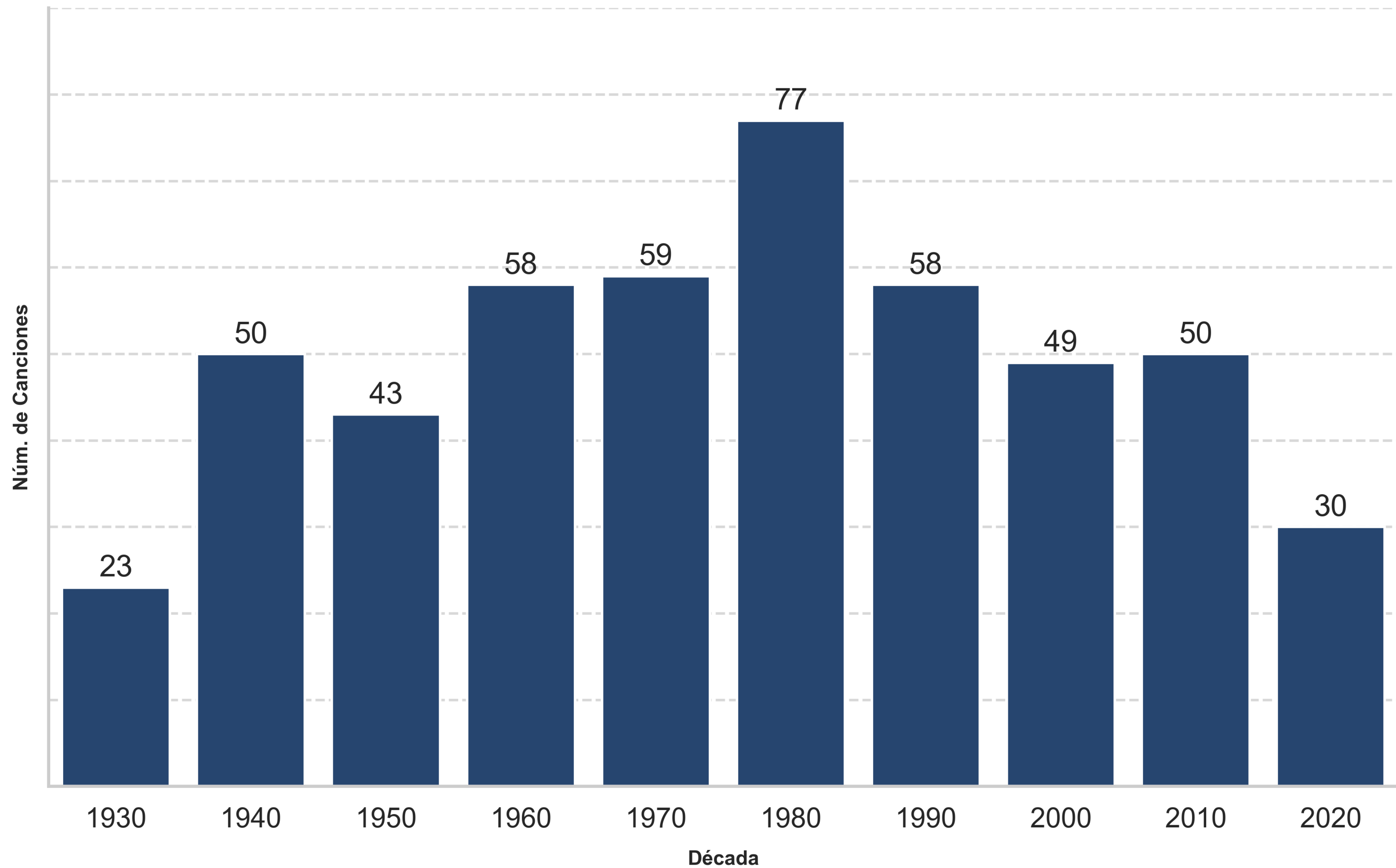
Exploración del Dataset



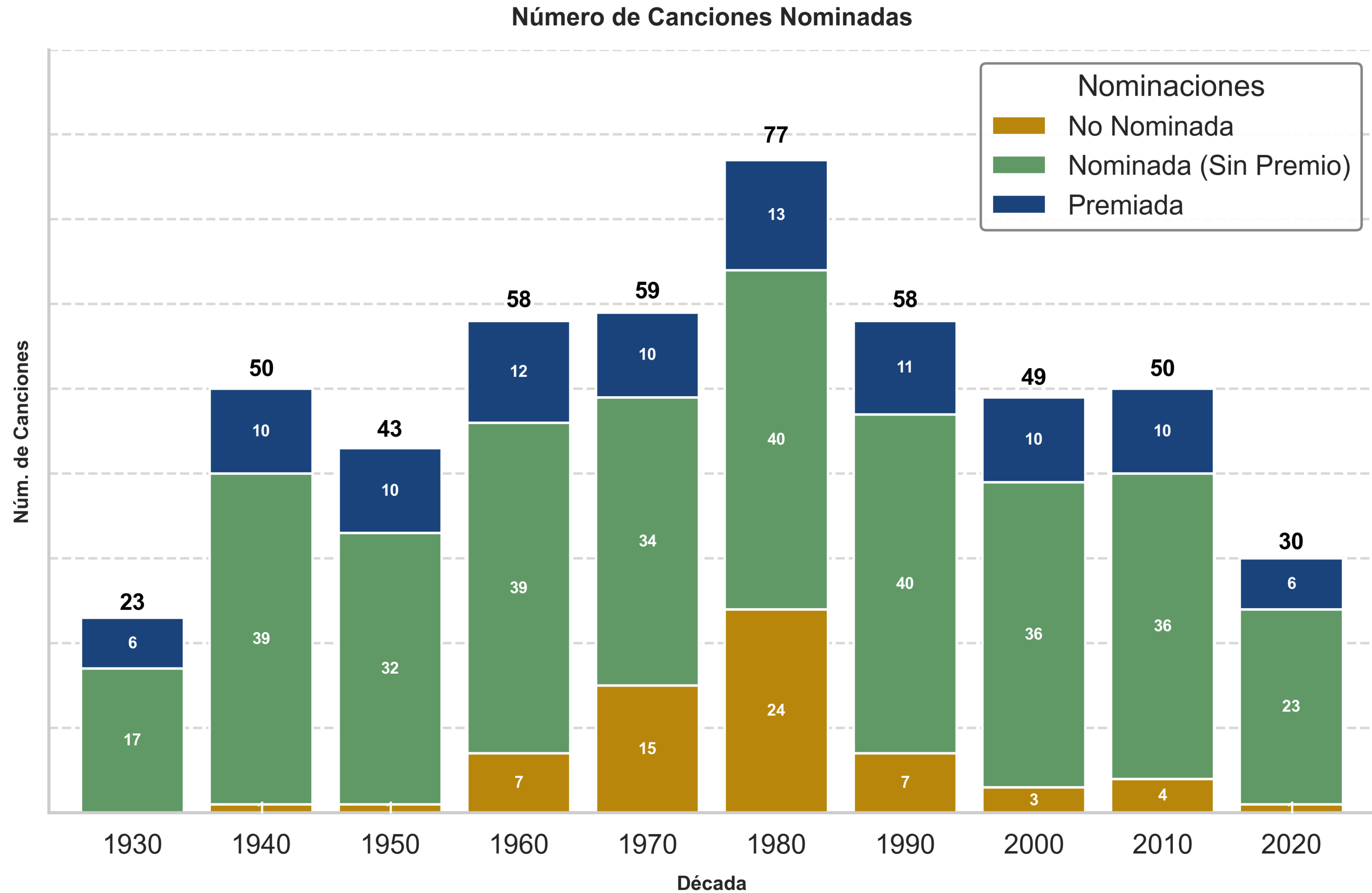
- 500 canciones
- 467 films
- 1937 -2025
- 602 artistas
- 2042 songwriters
- 196M streams (avg)
- 130 grabaciones (avg)
- 3'30" duración (avg)

Exploración del Dataset

Número de Canciones por Década

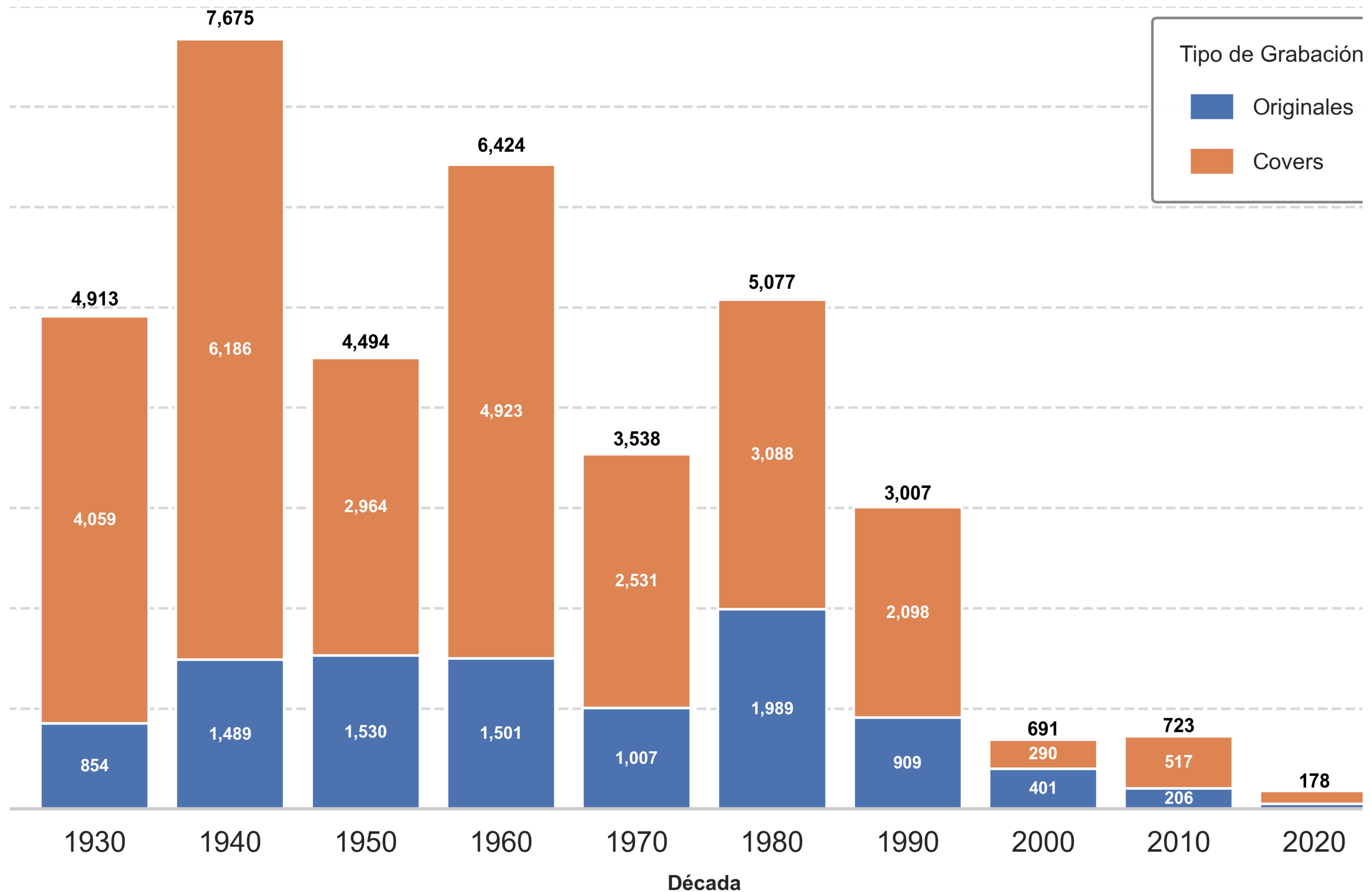


Exploración del Dataset

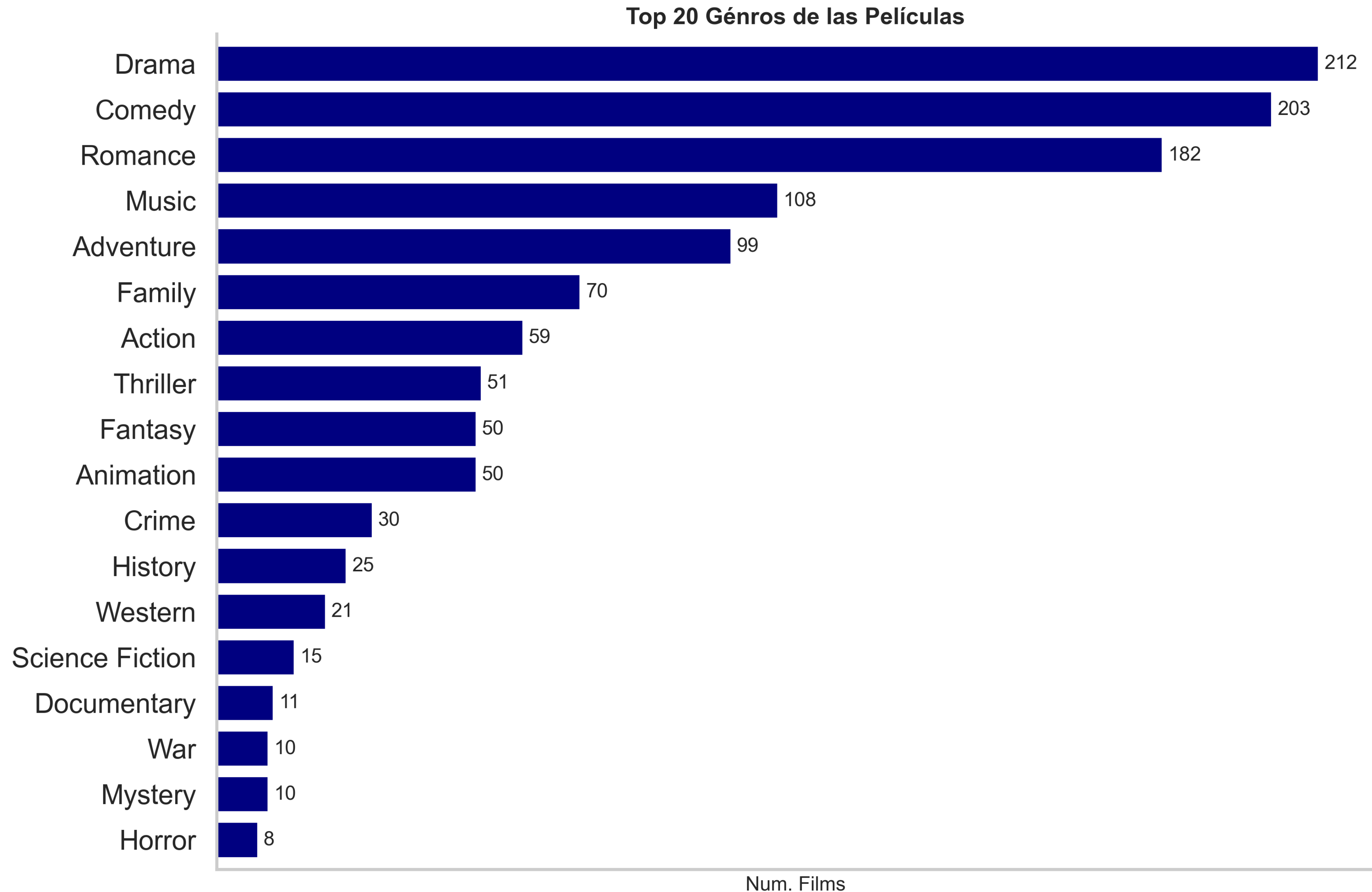


Exploración del Dataset

Número de Grabaciones: Originales vs. Covers



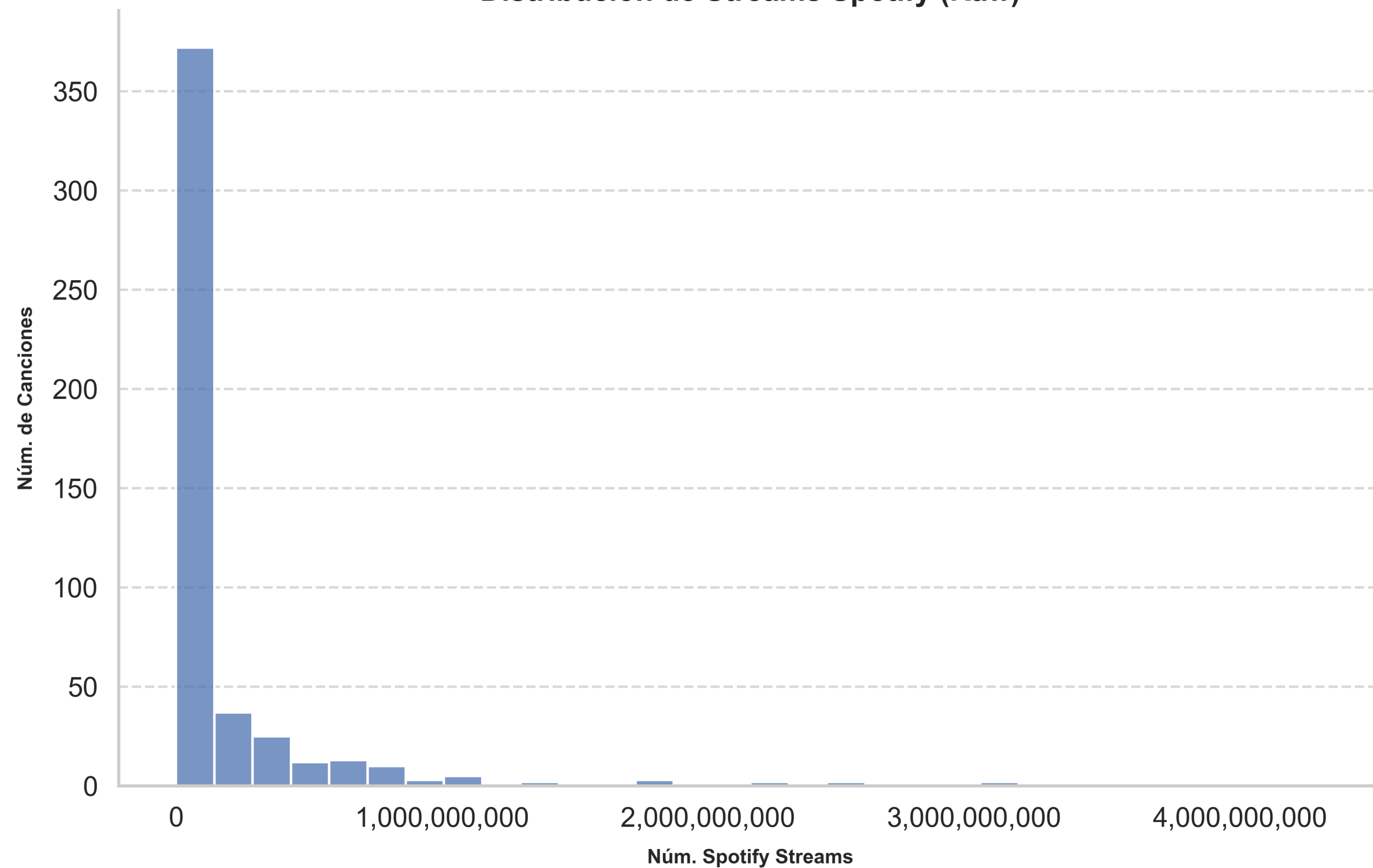
Exploración del Dataset (Films)



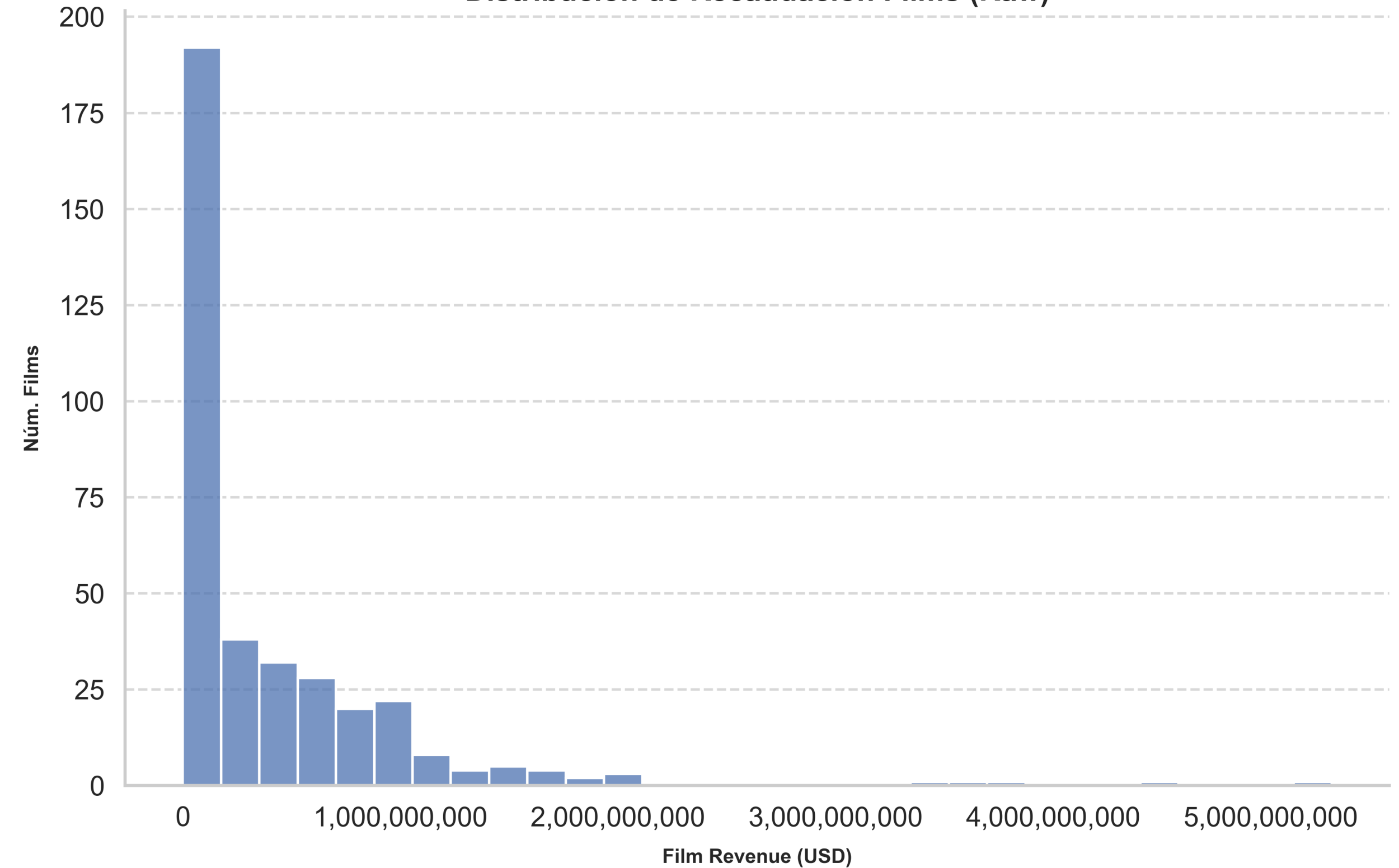
Streams / Taquilla Films

Cómo se distribuyen las reproducciones en Spotify y la recaudación de los films

Distribución de Streams Spotify (Raw)

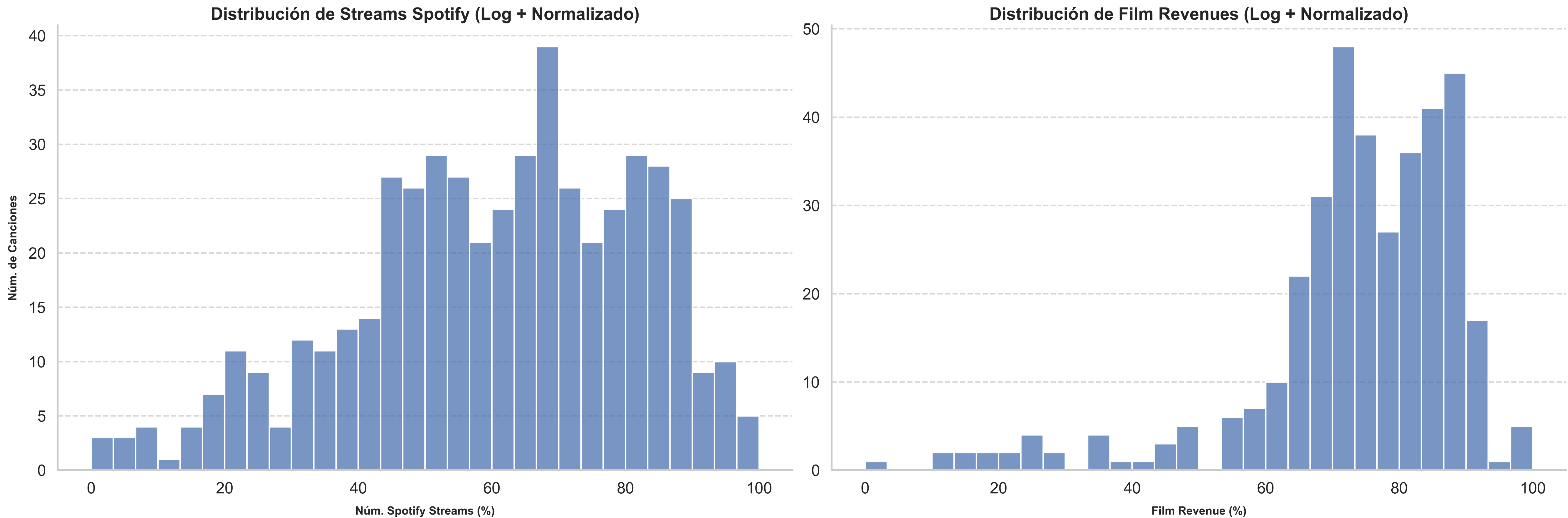


Distribución de Recaudación Films (Raw)



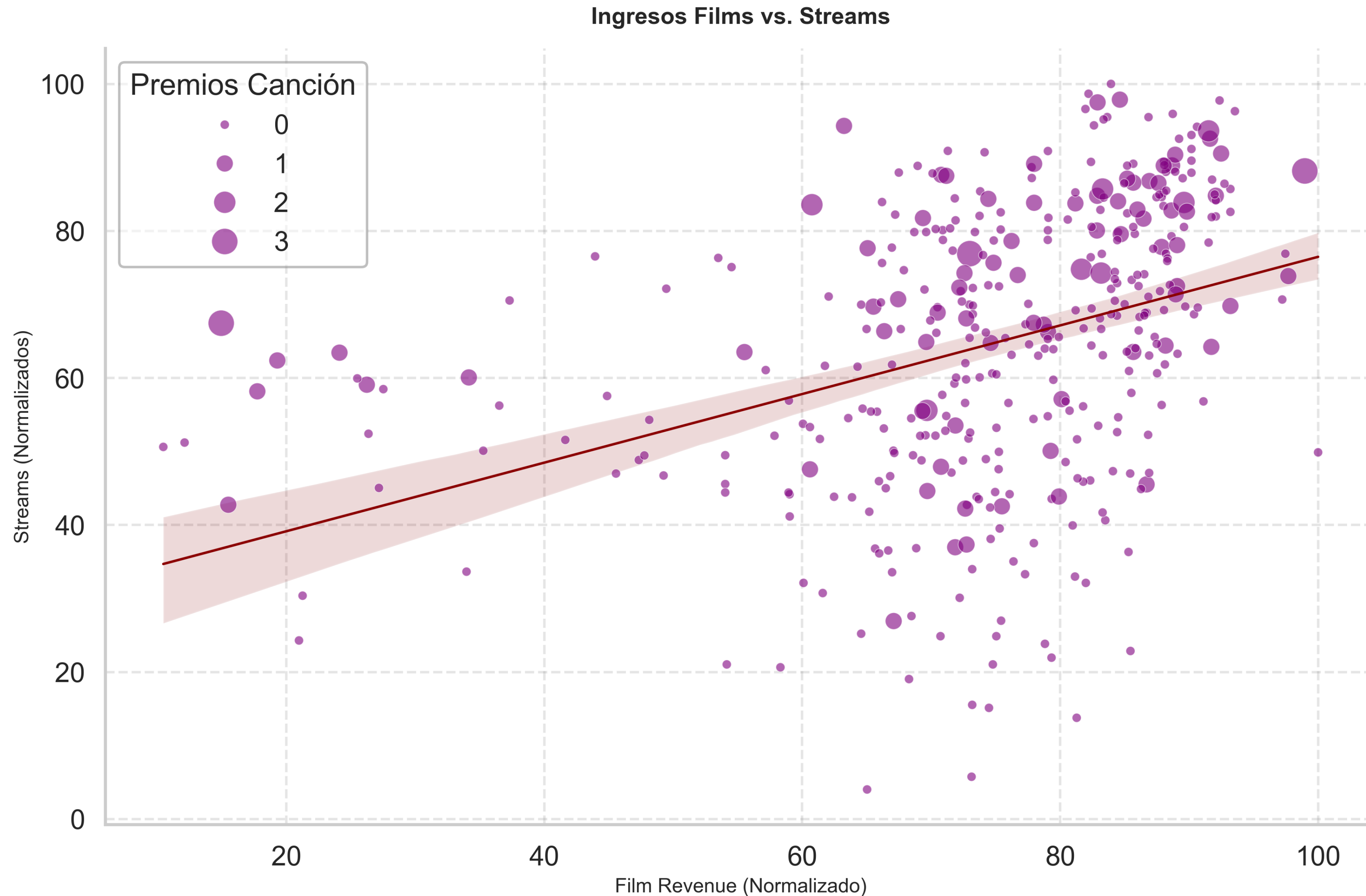
Streams / Taquilla Films

Cómo se distribuyen las reproducciones en Spotify y la recaudación de los films



Streams Spotify vs. Taquilla Films

¿Cómo se relaciona la recaudación de un film con el streaming de su canción original?



- Spearman: $r = 0,472$
- Índice determinación: aprox. $R^2 \approx 0,20$
- Relación positiva **leve**
- El valor de taquilla del film sólo refleja en un **20%** el nivel de streaming de la canción

Índice Multi-Dimensional

¿Qué factores contribuyen al éxito de una canción de película?

Factor Comercial

spotify_streams
album_sales

Factor Audiencia

lastfm_listeners
recordings

Factor Prestigio

awards_pct
acclaimed_score

Índice Multi-Dimensional

¿Qué factores contribuyen al éxito de una canción de película?

FÓRMULA BASE

 **Factor Comercial** } 35%

spotify_streams
album_sales

 **Factor Audiencia** } 35%

lastfm_listeners
recordings

 **Factor Prestigio** } 30%

awards_pct
acclaimed_score

Índice Multi-Dimensional

¿Qué factores contribuyen al éxito de una canción de película?

SESGO COMERCIAL

 **Factor Comercial** } **90%**
spotify_streams
album_sales

 **Factor Audiencia** } **5%**
lastfm_listeners
recordings

 **Factor Prestigio** } **5%**
awards_pct
acclaimed_score

FÓRMULA BASE

 **Factor Comercial** } **35%**
spotify_streams
album_sales

 **Factor Audiencia** } **35%**
lastfm_listeners
recordings

 **Factor Prestigio** } **30%**
awards_pct
acclaimed_score

SESGO PRESTIGIO

 **Factor Comercial** } **5%**
spotify_streams
album_sales

 **Factor Audiencia** } **5%**
lastfm_listeners
recordings

 **Factor Prestigio** } **90%**
awards_pct
acclaimed_score

Análisis de Sensibilidad - Top 25

¿Cuáles son las mejores canciones de película de todos los tiempos?

Top Songs by Commercial Score:			
	song_title	year_song	success_score_low
1	Lose Yourself	2002	95.45
2	Stayin' Alive	1977	91.66
3	My Heart Will Go On	1997	90.96
4	(Everything I Do) I Do It For You	1991	89.16
5	I Don't Want To Miss A Thing	1998	88.98
6	How Deep Is Your Love	1977	88.95
7	White Christmas	1942	88.47
8	Mrs. Robinson	1967	88.31
9	Knockin' On Heaven's Door	1973	88.06
10	Earned It	2015	87.66
11	I Just Called To Say I Love You	1984	87.58
12	Jailhouse Rock	1957	86.92
13	Skyfall	2012	86.46
14	Young And Beautiful	2013	86.35
15	More Than a Woman	1977	86.25
16	I Have Nothing	1992	86.23
17	Maniac	1983	86.02
18	Can't Stop The Feeling	2016	85.99
19	Against All Odds (Take A Look At Me Now)	1984	85.66
20	Streets Of Philadelphia	1993	85.56
21	Night Fever	1977	85.5
22	Because You Loved Me	1996	84.85
23	Tears In Heaven	1991	84.56
24	Help!	1965	84.31
25	Say You, Say Me	1985	84.09

Top Songs by Success Score:			
	song_title	year_song	song_success_score
1	My Heart Will Go On	1997	92.82
2	Lose Yourself	2002	88.05
3	Streets Of Philadelphia	1993	88.03
4	White Christmas	1942	84.5
5	I Just Called To Say I Love You	1984	83.87
6	The Way We Were	1973	80.82
7	(Everything I Do) I Do It For You	1991	80.19
8	Can You Feel The Love Tonight	1994	78.33
9	I Don't Want To Miss A Thing	1998	77.46
10	Tears In Heaven	1991	77.19
11	Skyfall	2012	76.0
12	Mrs. Robinson	1967	75.97
13	Stayin' Alive	1977	75.64
14	Moon River	1961	75.54
15	All The Way	1957	74.9
16	Say You, Say Me	1985	74.59
17	Over The Rainbow	1939	74.32
18	Flashdance...What A Feeling	1983	73.71
19	Maniac	1983	73.57
20	A Whole New World	1992	73.34
21	Shallow	2018	73.33
22	Take My Breath Away	1986	73.08
23	Arthur's Theme (Best That You Can Do)	1981	72.83
24	Raindrops Keep Fallin' On My Head	1969	72.77
25	Against All Odds (Take A Look At Me Now)	1984	72.45

Top Songs by Prestige Score:			
	song_title	year_song	success_score_high
1	My Heart Will Go On	1997	98.48
2	Streets Of Philadelphia	1993	90.36
3	Moon River	1961	86.92
4	Days Of Wine And Roses	1962	86.42
5	Lose Yourself	2002	82.59
6	The Way We Were	1973	81.48
7	Arthur's Theme (Best That You Can Do)	1981	78.82
8	A Whole New World	1992	78.79
9	The Shadow Of Your Smile	1965	77.97
10	What Was I Made For?	2023	77.86
11	Evergreen (Love Theme From A Star Is Born)	1976	77.5
12	I Just Called To Say I Love You	1984	73.7
13	Flashdance...What A Feeling	1983	72.38
14	Theme From Shaft	1971	72.25
15	Raindrops Keep Fallin' On My Head	1969	72.18
16	Shallow	2018	70.84
17	Beauty And The Beast	1991	70.81
18	Fame	1980	70.55
19	You Light Up My Life	1977	69.61
20	White Christmas	1942	67.16
21	Over The Rainbow	1939	65.33
22	Swinging On A Star	1944	64.72
23	Can You Feel The Love Tonight	1994	64.7
24	All The Way	1957	64.63
25	When You Wish Upon A Star	1940	64.36

Grandes Éxitos “Indiscutibles”

Consensus Blockbusters



Movie poster for **TITANIC**, featuring Leonardo DiCaprio and Kate Winslet. The poster shows the ship's bow and the couple's faces. Text on the poster includes: "NOTHING ON EARTH COULD COME BETWEEN THEM.", "LEONARDO DiCAPRIO KATE WINSLET", "A JAMES CAMERON FILM", "TITANIC", and "PARAMOUNT PICTURES PRESENTS A JAMES CAMERON FILM 'TITANIC' LEONARDO DiCAPRIO KATE WINSLET".

MY HEART WILL GO ON
“Titanic” (1997)

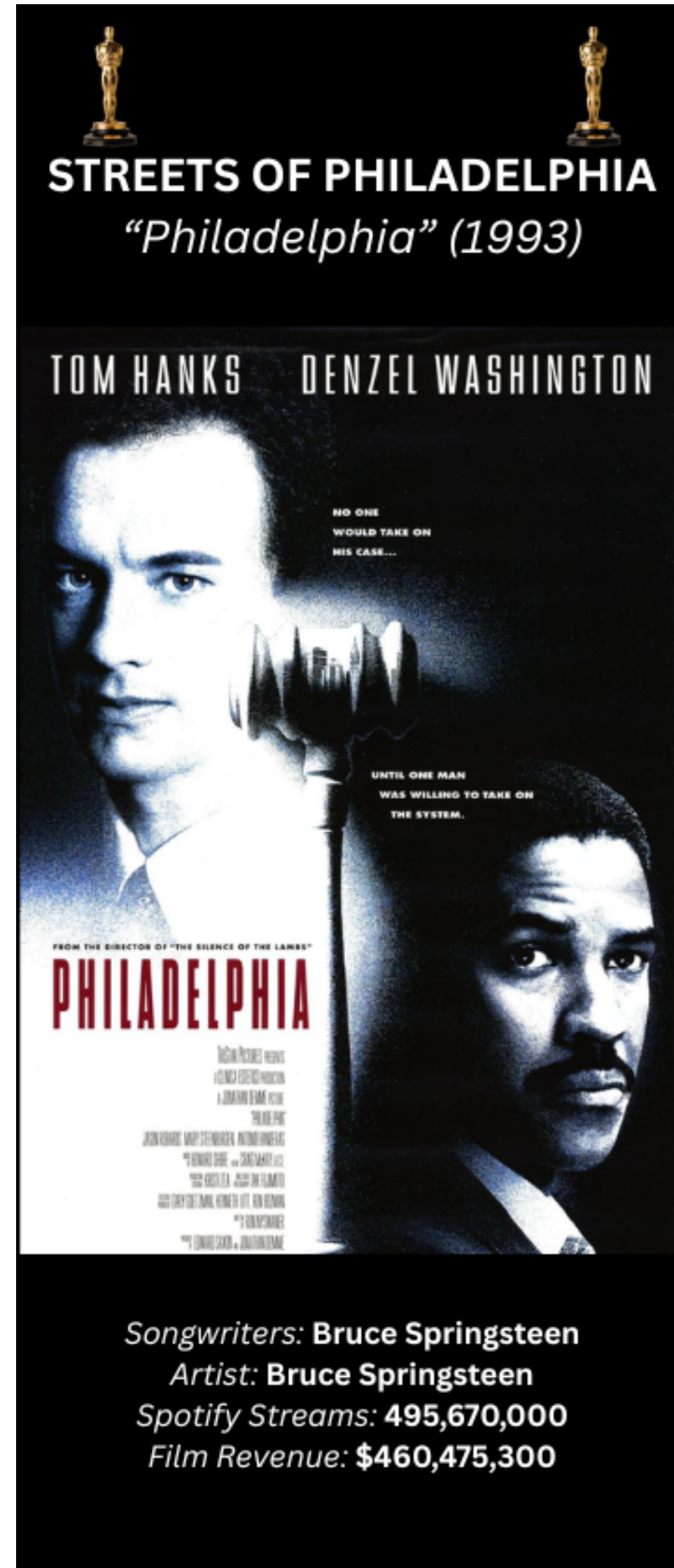
Songwriters: James Horner, Will Jennings
Artist: Céline Dion
Spotify Streams: 719,790,000
Film Revenue: \$4,541,627,542



Movie poster for **8 Mile**, featuring Eminem. The poster shows Eminem in a dark jacket and beanie, looking down. Text on the poster includes: "EMINEM KIM BASINGER BRITTANY MURPHY MEKHI PHIFER", "8 Mile", "Every moment is another chance", and "NOVEMBER 8".

LOSE YOURSELF
“Eight Mile” (2002)

Songwriters: Eminem, Jeff Bass, Luis Resto
Artist: Eminem
Spotify Streams: 2,969,600,000
Film Revenue: \$434,641,085



Movie poster for **PHILADELPHIA**, featuring Tom Hanks and Denzel Washington. The poster shows Hanks and Washington in suits, with a gavel in the foreground. Text on the poster includes: "TOM HANKS DENZEL WASHINGTON", "NO ONE WOULD TAKE ON HIS CASE...", "UNTIL ONE MAN WAS WILLING TO TAKE ON THE SYSTEM.", "FROM THE DIRECTOR OF 'THE SILENCE OF THE LAMBS'", "PHILADELPHIA", and "DOLBY DIGITAL".

STREETS OF PHILADELPHIA
“Philadelphia” (1993)

Songwriters: Bruce Springsteen
Artist: Bruce Springsteen
Spotify Streams: 495,670,000
Film Revenue: \$460,475,300



Movie poster for **HOLIDAY INN**, featuring Bing Crosby, Fred Astaire, and Marjorie Reynolds. The poster shows the three stars in a festive, musical setting. Text on the poster includes: "Irving Berlin's HOLIDAY INN", "A PARAMOUNT PICTURE", "Starring BING CROSBY FRED ASTAIRE with MARJORIE REYNOLDS VIRGINIA DALE-WALTER ABEL", "A MARK SANDRICH Production", "LYRICS AND MUSIC BY IRVING BERLIN", and "Screen Play by Claude Broyan".

WHITE CHRISTMAS
“Holiday Inn” (1942)

Songwriters: Irving Berlin
Artist: Bing Crosby
Spotify Streams: 659,600,000
Film Revenue: \$74,066,641



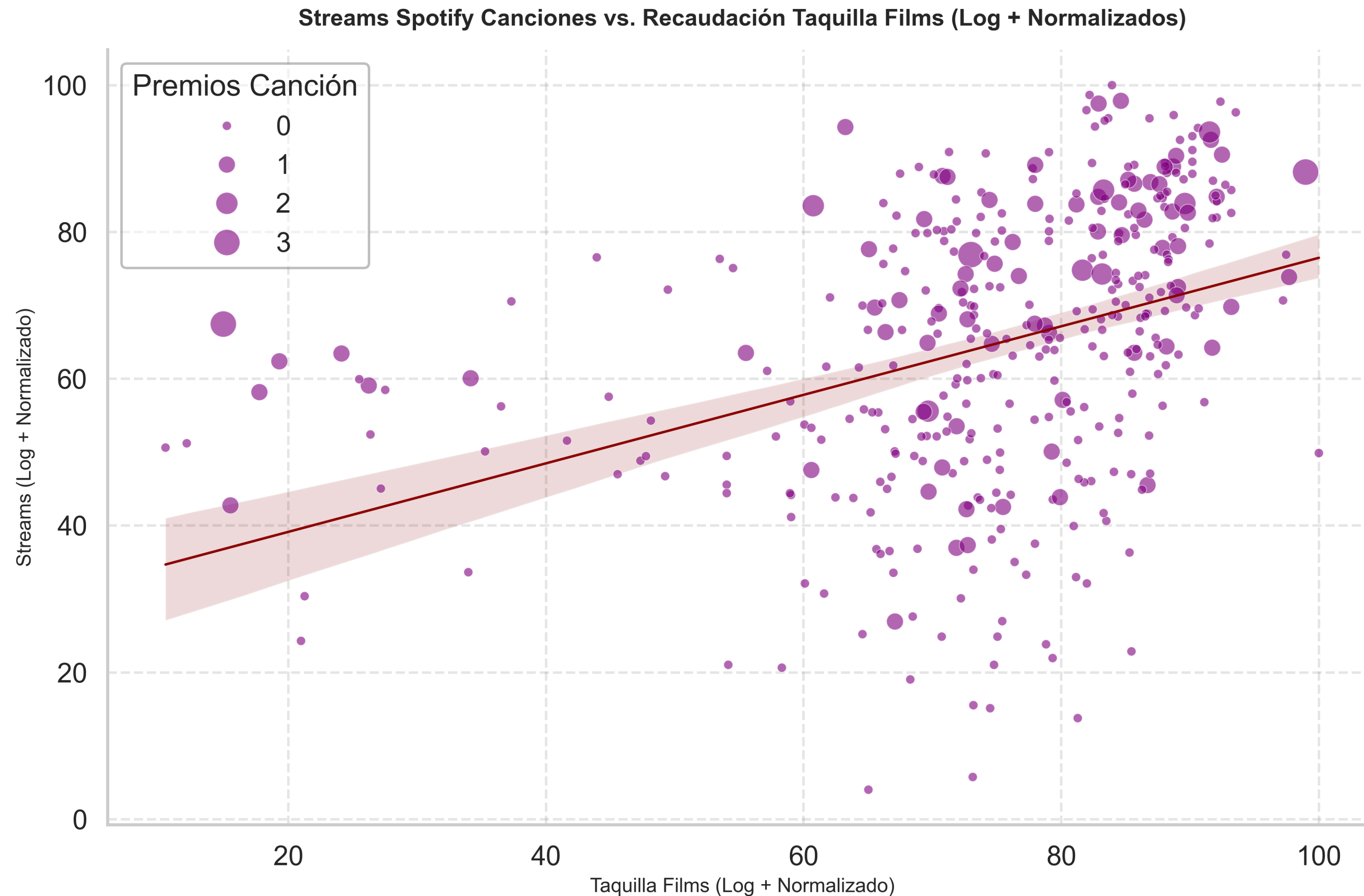
Movie poster for **The Woman in Red**, featuring Gene Wilder and Charles Grodin. The poster shows a woman in a red dress and a man peeking over a ledge. Text on the poster includes: "I JUST CALLED TO SAY I LOVE YOU", "“The Woman In Red” (1984)", "The Woman in Red", "Shy, quiet Teddy Pierce wanted a little adventure. And one day it walked into his life in a red silk dress. Now his wife is packing a gun. His friends are going nuts trying to cover for him and he's about to get caught with his pants down on the six o'clock news. Be very careful of what you want... Because you just might get it.", "GENE WILDER CHARLES GRODIN JOSEPH BOLOGNA JUDITH FINEY MICHAEL BIEBERGELSON KELLY LE BROCK GILDA RAINER", "A VICTOR DRAI PRODUCTION 'THE WOMAN IN RED' WITH JOHN MORRIS STEVE WONDER", "STARRING STEVE WONDER AND DIANNE WARWICK JOHNNY FREED SCHILLER NORTON JACK FREEST SANDERS", "IN ASSOCIATION WITH JOHNNY FREED SCHILLER'S JEAN LOUP DABOIS AND YVES ROBERT PRODUCED BY VICTOR DRAI", "DOLBY DIGITAL", "THE WOMAN IN RED", "A VICTOR DRAI PRODUCTION", "GENE WILDER CHARLES GRODIN JOSEPH BOLOGNA JUDITH FINEY MICHAEL BIEBERGELSON KELLY LE BROCK GILDA RAINER", "STARRING STEVE WONDER AND DIANNE WARWICK JOHNNY FREED SCHILLER NORTON JACK FREEST SANDERS", "IN ASSOCIATION WITH JOHNNY FREED SCHILLER'S JEAN LOUP DABOIS AND YVES ROBERT PRODUCED BY VICTOR DRAI", "DOLBY DIGITAL", "THE WOMAN IN RED", "A VICTOR DRAI PRODUCTION".

I JUST CALLED TO SAY I LOVE YOU
“The Woman In Red” (1984)

Songwriters: Stevie Wonder
Artist: Stevie Wonder
Spotify Streams: 654,350,000
Film Revenue: \$78,394,205

Éxito Canción vs. Éxito Taquilla

¿Cómo se relaciona el éxito de una canción de película con la taquilla del film?

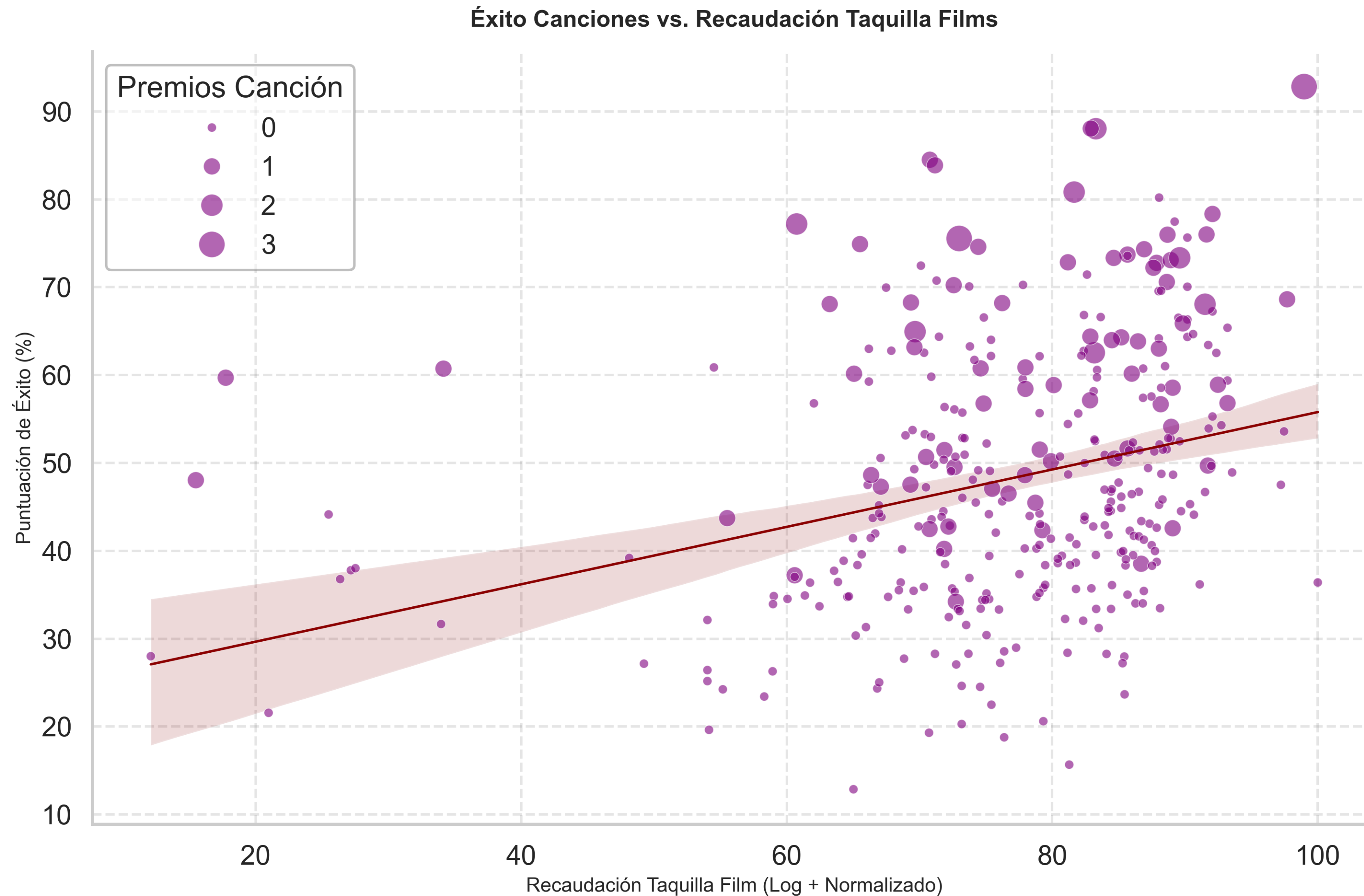


$$r = 0,47$$

$$R^2 \approx 20\%$$

Éxito Canción vs. Éxito Taquilla

¿Cómo se relaciona el éxito de una canción de película con la taquilla del film?



$$r = 0,47$$

$$R^2 \approx 20\%$$

$$r = 0,34$$

$$R^2 \approx 12\%$$

CONCLUSIONES

- El **éxito** de una canción original de película está poco relacionado con la recaudación del film
- Las mejores canciones de película de todos los tiempos son aquellas que logran el **consenso** en diferentes escenarios (valorando factores como el **consumo**, el **engagement** y el **prestigio**)

PROPUESTAS

- Ampliar el dataset (**1000** canciones)
- Añadir variables de **aspectos musicales** (bpm, tonalidad...)
- Indagar en los **repartos** de derechos de autor de los compositores
- Incorporar también una puntuación del **éxito de las películas**
- Optimizar el modelo mediante algoritmos de **machine learning**



Agradecimientos: Irene, Joan, Alana, Dago, Yunier, IT Academy, Bcn Activa



**“Always look on the
bright side of life”**

(Monty Python, 1979)



The End